

Solution Requirements

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system must provide secure single sign-on (SSO) and multi-factor authentication (MFA) via OAuth 2.0 or standard encrypted credential pathways for Farmers, Agronomists, and Admins to safely access corporate dashboards and custom data entry layers.	High
2	Authorization levels	Implement Role-Based Access Control (RBAC) defining three primary tiers: 'Farmers' can read/write own farm telemetry; 'Agronomists' possess access to region-wide filtered datasets and advisory tools; 'Admins' exercise full operational control over configurations and systemic integrations.	High
3	External interfaces	The application must interface seamlessly with global meteorological APIs (e.g., OpenWeatherMap) to fetch live local micro-climate metrics, and consume agricultural soil maps via RESTful web endpoints.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	Transactions processing	The platform must handle high-throughput telemetry data packet streams transmitted from field-level IoT sensors without dropping connections, ensuring records are parsed and committed atomic-style to the central transaction log.	High
5	Reporting	Provide automated PDF crop performance reporting sheets and interactive web dashboards showcasing historical matrix patterns, moisture curves, yield predictions, and recommended structural interventions.	Medium
6	Business rules, etc.	If detected Nitrogen-Phosphorus-Potassium (NPK) ratios deviate more than 20% from optimal crop baselines, trigger immediate automated warnings to the corresponding farmer profiles.	High
7	Compliance to laws or regulations	Data storage, localization architecture, and user privacy protection mechanism layouts must comply completely with standard regional information rules and global data protection guidelines.	High
8	Other	Provide comprehensive off-line operational queues that cache manual farm telemetry changes locally when mobile networks drop, syncing flawlessly once connections reappear.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System response times for loading interactive telemetry visualizations and spatial map views must be fully optimized for low-latency delivery.	<i>Dashboard analytics panels and page assets load completely within < 2 seconds under normal bandwidth.</i>
2	Scalability	The dynamic cloud architecture must smoothly scale computing allocation as additional field sensors and accounts register.	<i>Effortlessly supports concurrent tracking pipelines up to 10,000 active concurrent users/streams.</i>

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	All resting telemetry records and active application networking routes must utilize robust mathematical security patterns.	<i>Enforced end-to-end AES-256 encryption on structural databases and TLS 1.3 for active API channels.</i>
4	Reliability & Availability	The production operational status of web environments and analytical microservices must maintain consistent runtimes.	<i>Guaranteed minimum of 99.9% application uptime per calendar month.</i>
5	Usability & Accessibility	User dashboard viewports and form elements must accommodate cross-device responsiveness and multi-accessible guidelines.	<i>Full structural compliance with WCAG 2.1 AA accessibility interface layouts.</i>
6	Other	System code architectures, deployment steps, and microservices must remain structurally modular to facilitate quick extensions.	<i>Maintainability: Clean codebase documentation allowing new platform feature integrations within standard sprint windows.</i>